Pose Estimation

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Use Python to demonstrate **Ultralytics**: Pose Estimation effects on images, videos, and real-time detection.

1. Model Introduction

Pose estimation is a task that involves identifying the location of specific points in an image (usually called keypoints). Keypoints can represent various parts of an object, such as joints, landmarks, or other significant features. The position of keypoints is typically represented by a set of 2D `[x, y]` or 3D `[x, y, visible]` coordinates.

The output of a pose estimation model is a set of points representing the keypoints of objects in the image, usually including confidence scores for each point. When you need to identify specific parts of objects in a scene and their positional relationships to each other, pose estimation is a good choice.

In the default pose model of YOLO26, there are 17 keypoints, each representing different parts of the human body. Here is the mapping of each index to the corresponding body joint:

1. Nose
2. Left eye
3. Right eye
4. Left ear
5. Right ear
6. Left shoulder
7. Right shoulder
8. Left elbow
9. Right elbow
10. Left wrist
11. Right wrist
12. Left hip
13. Right hip
14. Left knee
15. Right knee
16. Left ankle
17. Right ankle

2. Pose Estimation: Image

Use yolo26n-pose_ncnn_model to predict images in the ultralytics26 folder (not built-in ultralytics images).

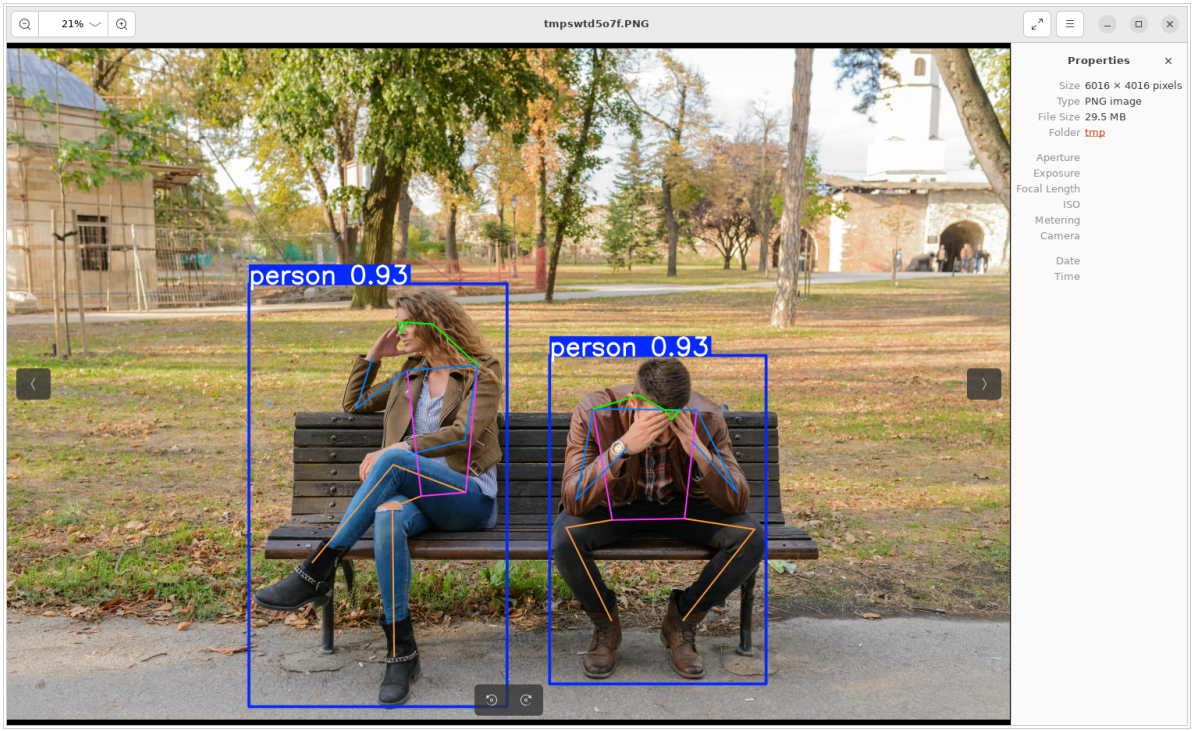
Enter the code folder:

```
cd /home/jetson/ultralytics/yahboom_demo
```

Run the code:

```
python3 03.pose_image.py
```

Effect Preview



Example code:

```
from ultralytics import YOLO

# Load a model
model = YOLO("/home/jetson/ultralytics/yolo26n-pose.engine")  # pretrained
yolo26n-pose model

# Run batched inference on a list of images
results = model("/home/jetson/ultralytics/ultralytics/assets/people.jpg")  #
return a list of Results objects

# Process results list
for result in results:
    # boxes = result.boxes  # Boxes object for bounding box outputs
    # masks = result.masks  # Masks object for segmentation masks outputs
    keypoints = result.keypoints  # Keypoints object for pose outputs
    # probs = result.probs  # Probs object for classification outputs
    # obb = result.obb  # Oriented boxes object for OBB outputs
    result.show()  # display to screen
```

```
    result.save(filename="/home/jetson/ultralytics/output/people_output.jpg")  # save to disk
```

3. Pose Estimation: Video

Use yolo26n-pose_ncnn_model to predict videos in the ultralytics26 folder (not built-in ultralytics videos).

Enter the code folder:

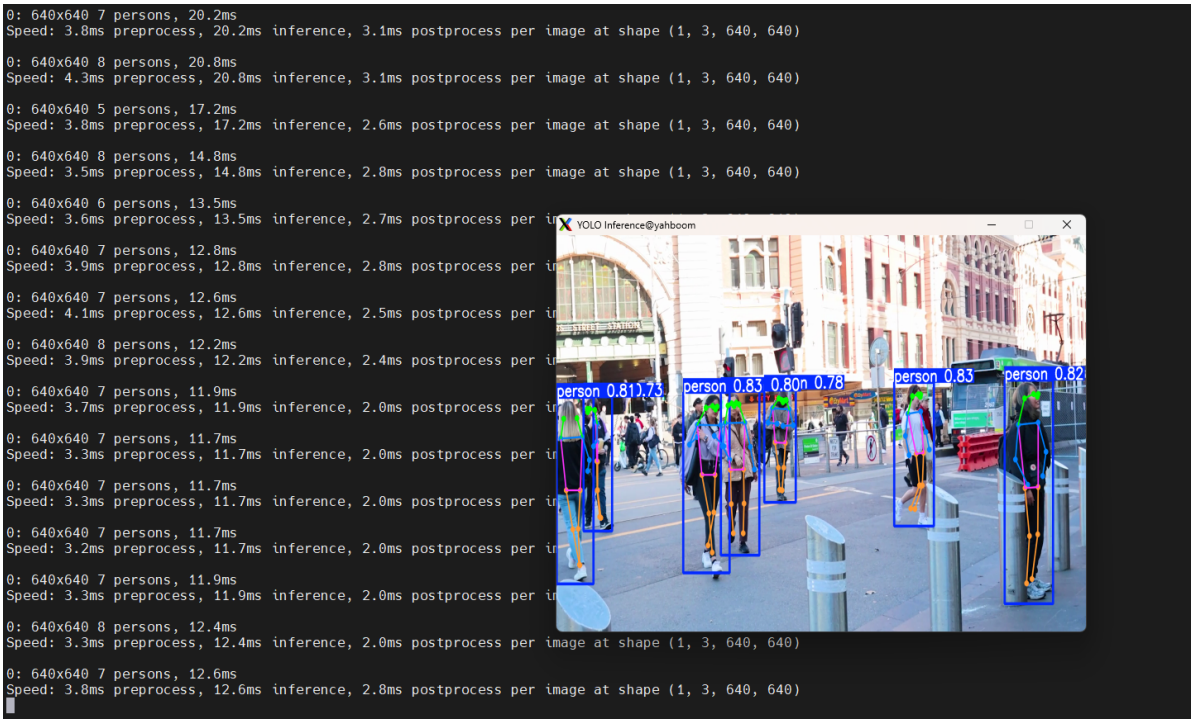
```
cd /home/jetson/ultralytics/yahboom_demo
```

Run the code:

```
python3 03.pose_video.py
```

Effect Preview

yolo recognition output video location: /home/jetson/ultralytics/output/



Example code:

```
import cv2
from ultralytics import YOLO

# Load the YOLO model
model = YOLO("/home/jetson/ultralytics/yolo26n-pose.engine")

# Open the video file
video_path = "/home/jetson/ultralytics/ultralytics/videos/people.mp4"
cap = cv2.VideoCapture(video_path)

# Get the video frame size and frame rate
frame_width = int(cap.get(cv2.CAP_PROP_FRAME_WIDTH))
frame_height = int(cap.get(cv2.CAP_PROP_FRAME_HEIGHT))
```

```
fps = int(cap.get(cv2.CAP_PROP_FPS))

# Define the codec and create a VideoWriter object to output the processed video
output_path = "/home/jetson/ultralytics/output/03.people_animals_output.mp4"
fourcc = cv2.VideoWriter_fourcc(*'mp4v')  # You can use 'XVID' or 'mp4v'
depending on your platform
out = cv2.VideoWriter(output_path, fourcc, fps, (frame_width, frame_height))

# Loop through the video frames
while cap.isOpened():
    # Read a frame from the video
    success, frame = cap.read()

    if success:
        # Run YOLO inference on the frame
        results = model(frame)

        # Visualize the results on the frame
        annotated_frame = results[0].plot()

        # Write the annotated frame to the output video file
        out.write(annotated_frame)

        # Display the annotated frame
        cv2.imshow("YOLO Inference", cv2.resize(annotated_frame, (640, 480)))

        # Break the loop if 'q' is pressed
        if cv2.waitKey(1) & 0xFF == ord("q"):
            break
    else:
        # Break the loop if the end of the video is reached
        break
# Release the video capture and writer objects, and close the display window
cap.release()
out.release()
cv2.destroyAllWindows()
```

4. Pose Estimation: Real-time Detection

4.1. USB Camera

Use yolo26n-pose_ncnn_model to predict USB camera footage.

Enter the code folder:

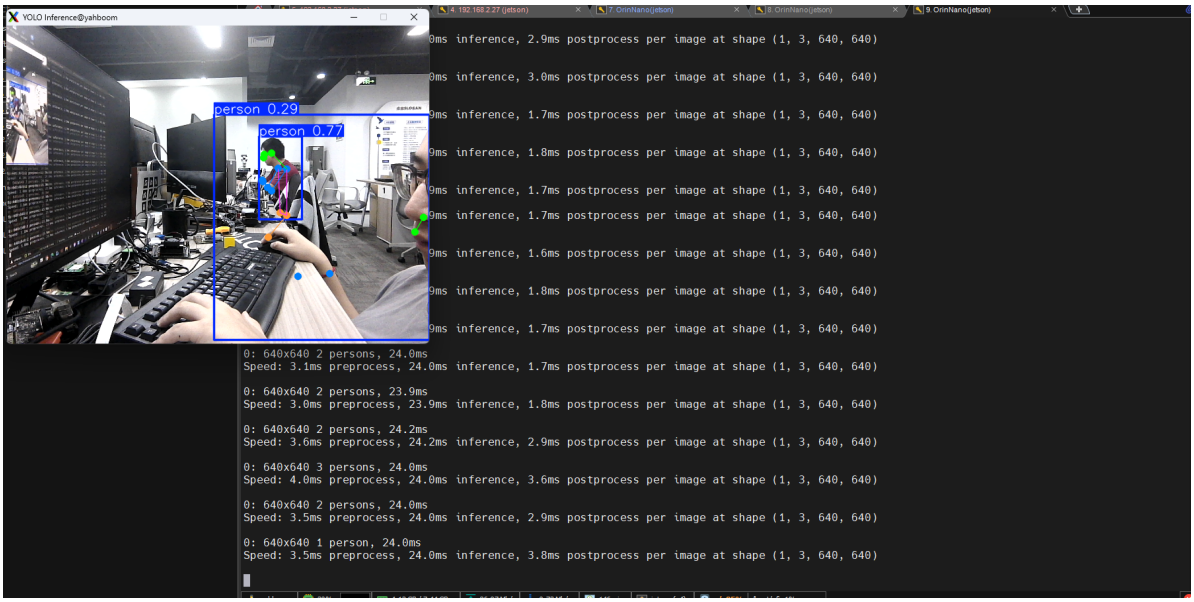
```
cd /home/jetson/ultralytics/yahboom_demo
```

Run the code: Click on the preview screen, press q key to terminate the program!

```
python3 03.pose_camera_usb.py
```

Effect Preview

yolo recognition output video location: /home/jetson/ultralytics/output/



Example code:

```
import cv2
from ultralytics import YOLO

# Load the YOLO model
model = YOLO("/home/jetson/ultralytics/yolo26n-pose.engine")

# Open the cammera
cap = cv2.VideoCapture(0)
cap.set(6, cv2.VideoWriter.fourcc('M', 'J', 'P', 'G'))
cap.set(cv2.CAP_PROP_FRAME_WIDTH, 640)
cap.set(cv2.CAP_PROP_FRAME_HEIGHT, 480)

# Get the video frame size and frame rate
frame_width = int(cap.get(cv2.CAP_PROP_FRAME_WIDTH))
frame_height = int(cap.get(cv2.CAP_PROP_FRAME_HEIGHT))
fps = int(cap.get(cv2.CAP_PROP_FPS))

# Define the codec and create a VideoWriter object to output the processed video
output_path = "/home/jetson/ultralytics/output/03.pose_camera_usb.mp4"
fourcc = cv2.VideoWriter_fourcc(*'mp4v')  # You can use 'XVID' or 'mp4v'
depending on your platform
out = cv2.VideoWriter(output_path, fourcc, fps, (frame_width, frame_height))

# Loop through the video frames
while cap.isOpened():
    # Read a frame from the video
    success, frame = cap.read()

    if success:
        # Run YOLO inference on the frame
        results = model(frame)

        # Visualize the results on the frame
        annotated_frame = results[0].plot()
```

```
        # Write the annotated frame to the output video file
        out.write(annotated_frame)

        # Display the annotated frame
        cv2.imshow("YOLO Inference", cv2.resize(annotated_frame, (640, 480)))

        # Break the loop if 'q' is pressed
        if cv2.waitKey(1) & 0xFF == ord("q"):
            break
    else:
        # Break the loop if the end of the video is reached
        break
# Release the video capture and writer objects, and close the display window
cap.release()
out.release()
cv2.destroyAllWindows()
```

4.2. CSI Camera

Use yolo26n-pose_ncnn_model to predict CSI camera footage.

Enter the code folder:

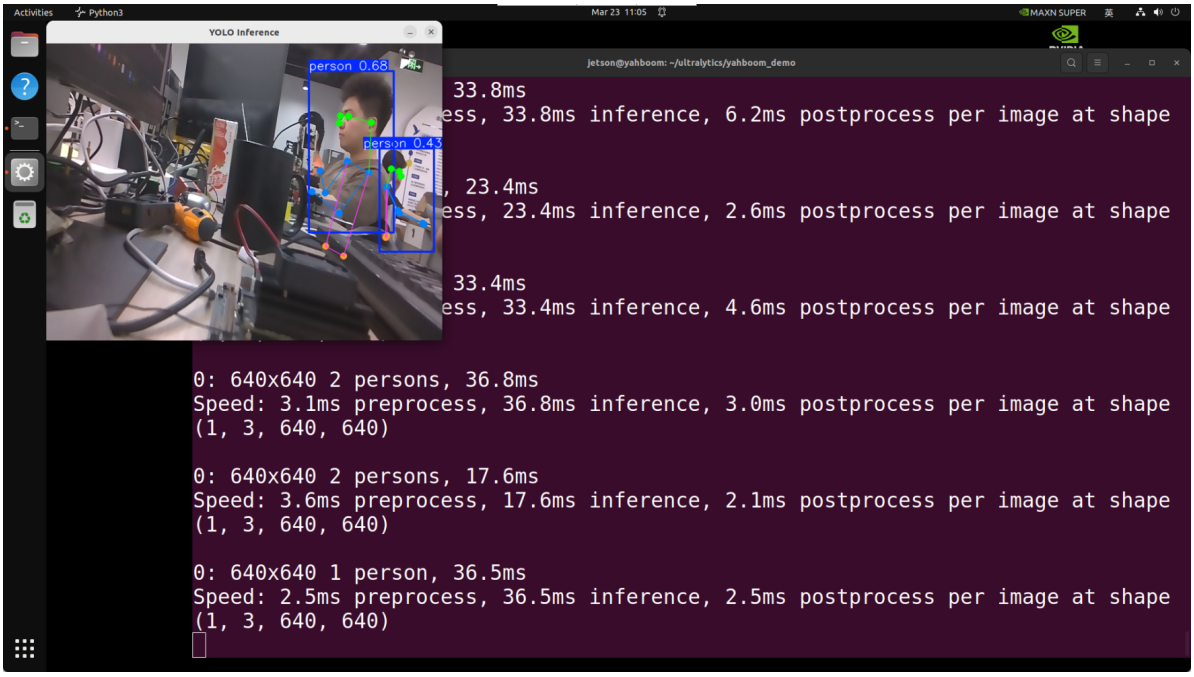
```
cd /home/jetson/ultralytics/yahboom_demo
```

Run the code: Click on the preview screen, press q key to terminate the program!

```
python3 03.pose_camera_csi.py
```

Effect Preview

yolo recognition output video location: /home/jetson/ultralytics/output/



Example code:

```
import cv2
```

```python
from ultralytics import YOLO
from jetcam.csi_camera import CSICamera

# Load the YOLO model
model = YOLO("/home/jetson/ultralytics/yolo26n-pose.engine")

# Open the camera (CSI Camera)
cap = CSICamera(capture_device=0, width=640, height=480)

# Get the video frame size and frame rate
frame_width = 640
frame_height = 480
fps = 30

# Define the codec and create a VideoWriter object to output the processed video
output_path = "/home/jetson/ultralytics/output/03.pose_camera_csi.mp4"
fourcc = cv2.VideoWriter_fourcc(*'mp4v')  # You can use 'XVID' or 'mp4v'
depending on your platform
out = cv2.VideoWriter(output_path, fourcc, fps, (frame_width, frame_height))

# Loop through the video frames
while True:
    # Read a frame from the camera
    frame = cap.read()

    if frame is not None:
        # Run YOLO inference on the frame
        results = model(frame)

        # Visualize the results on the frame
        annotated_frame = results[0].plot()

        # Write the annotated frame to the output video file
        out.write(annotated_frame)

        # Display the annotated frame
        cv2.imshow("YOLO Inference", cv2.resize(annotated_frame, (640, 480)))

        # Break the loop if 'q' is pressed
        if cv2.waitKey(1) & 0xFF == ord("q"):
            break
    else:
        # Break the loop if no frame is received (camera error or end of stream)
        print("No frame received, breaking the loop.")
        break

# Release the video capture and writer objects, and close the display window
cap.release()
out.release()
cv2.destroyAllWindows()
```

Reference Materials

[Pose Estimation - Ultralytics YOLO Documentation]